



Renewable Energy.....Naturally



Activity Areas: G1 Energy Solutions helps harness the Power of the SUN providing services directly and through its sister /partner concerns in areas of –

- **Grid / Utility Scale Power** of 100 KW to 1 MW / 10 MW or above evacuating at 11KV/33KV/66 KV
- **Off Grid Power** for Offices, Industry, Institutions upto 100 KW, LED lights, Telecom
- **Thermal / Fluid / Air heating** Solutions for Institutions, and Industry Either of **Owners Engineer OR Consultancy, design, engineering OR concept (project development) to commissioning through partner EPC teams ON turnkey basis**

Today's Situation – India

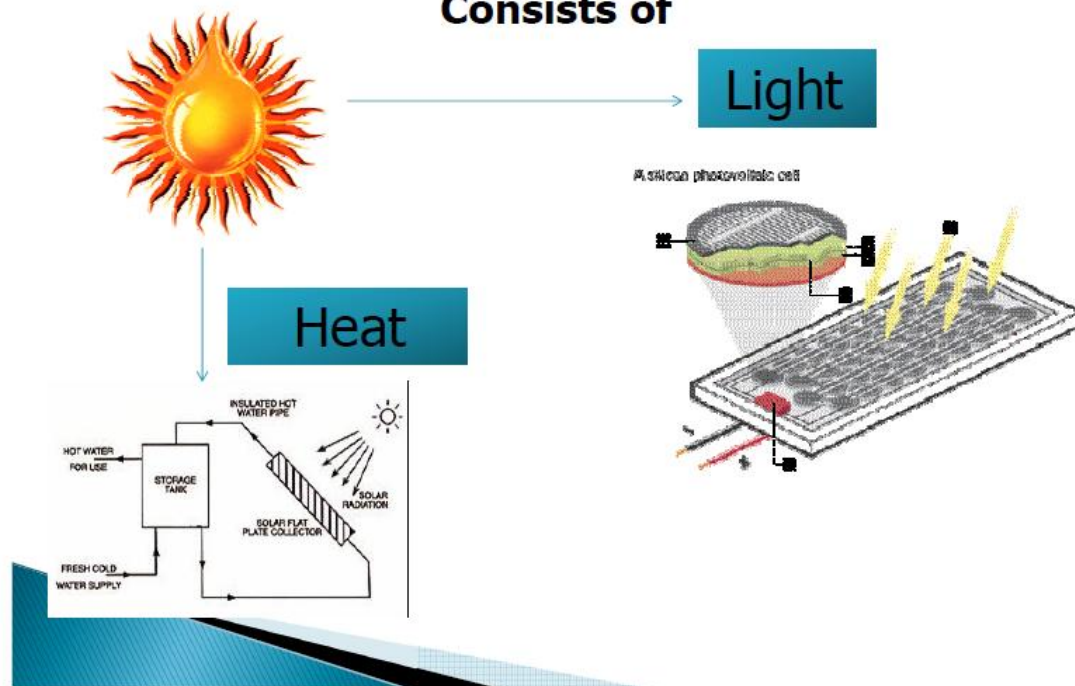
Power crisis in many developed states as TN, AP, and Maharashtra apart from other smaller states: factories run on diesel not on electricity!

- Many factories consume 70,000 to 90,000 Litres of diesel per month viz. (Approx. 30 – 50 Lac per month!) at present prices in TN, AP
- Average cost of diesel (incl. of transportation and maintenance) = Rs 19 to 21/- per unit. Further Diesel prices expected to go up escalating this problem.

- Based on above, the weighted Average electricity cost now for many industries and institutions is (EB & 25%DG) = Rs. 13 per unit
- New conventional power sources of electricity not on anvil.
- Hydro power suspect in India.
- Current coal prices are Rs. 5/Kg on regular e-auctions for industries and sometimes Rs. 8/Kg when supply is tight. Prices will only go up!
- Solar Only energy source close to grid parity now and can be own source of electricity for years to come!! Only source in YOUR HANDS

SOLAR ENERGY

Consists of



Project Management

Experienced Team, strong industry and technical credentials

- Management – Expertise in solar industry 40+ years, with project management expertise, both grid and off grid.
- Management team have installed together over 40 MW of grid connected solar power plants in Gujarat, Karnataka etc. under central government or state govt schemes in India evacuating at 11 KV / 33 KV or 66 KV in solar and 132 / 220 KV in Hydro power.
- Solar Project Managers- hands on experience of solar plants of KW to MW.
- Solar Design Engineers and Managers -Expertise on DC sizing, module sizing, loss calculations, equipment sizing, junction box designs, cable sizing.
- Field engineers and Managers / Supervisors – Engineers / Diplomas / ITI's - Trained for DC side in off grid and grid on solar power plants
- After Sales Support – Trained hands on equipment for operation, fault detection, measurement, data management, communication, safety, vendor and customer interaction etc.
- Equipment / Companies associated with: ABB, BHEL, Xantrex, Effekta, Bonfiglioli, Sungrow, SMA, Fronius, Steca, Outback, Studer, OPS, DB, etc.
- Off Grid installations done for various Govt., industry and institutions.

General Areas

Team Technical Track Record - Design and Engineering

- Executed 1 MW to 5 MW single systems with thin film, crystalline and CPV.
- Total of 40 MW plus installed capacities in India with evacuation at 11/33/66KV
- Mechanised and quick Civil foundations of various types as piling, hammering, open casting etc.
- JCB's, mixers, handling equip with 100 crew, with backup generators, shelters , lights etc. with statutory compliances and insurance.
- In-house certified structural engineer, civil engineer and associates

- Cable laying using armoured cables for underground laying with specs as per Electricity board rules
- DC Cable vendor sourcing, QC checks, cable loss calculations etc.
- Lightning arrestors, earth mat layout done with internal design using charcoal or chemical based units for array, inverter or substation areas.
- Multiple Designs done for 1 MW, 2 MW, 5 MW, 10 MW, 40 MW with crystalline and Thin Film or concentrators.
- Design done for CPV and HCPV for 1 MW and 2 MW.
- Designs for various off grid plants from 1 KW to 500 KW

PV Modules, Structures, Clamps

- Worked with different modules including sourcing for mono, multi – crystalline modules India and abroad of efficiencies up to 18.5%, Thin Film modules (a-Si) technology at 9 to 10% efficiencies, CPV with efficiencies of 22 to 26% and HCPV of 30% +
- **Handling equipment** – automated material handling at site.
- **Support frame:** MS, Prefab in-house /at site with galvanizing With material handling equipment / aux or DG power.
- **Module cleaning:** Using water spray and squeezer or Compressed air method either manual or automated.

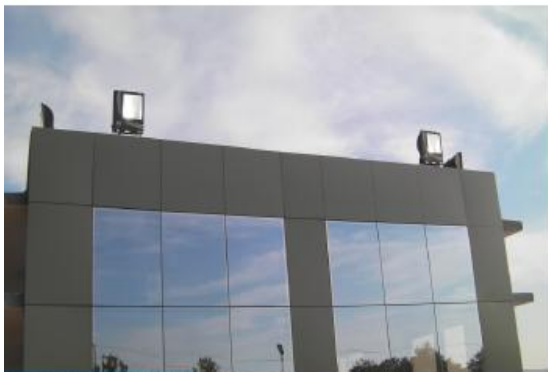
Few Projects operating and commissioned – Team Track Record

- 1 MW (2010–11), Crystalline, Thin Film, concentrator combined technology, evacuated at 11 KV to grid, Gujarat.
- 3 MW (2009),Crystalline, evacuated to grid 11 KV, Karnataka
- 3 MW (2011),Crystalline, evacuated to grid 11 KV, Karnataka
- 3 MW (2011),Crystalline, evacuated to grid 11 KV, Karnataka
- 3 MW (2012),Crystalline, evacuated to grid 11 KV, Karnataka
- 5 MW (2012),Crystalline, evacuated to grid 66 KV, Karnataka
- 7.5 MW (2012),Thin film, evacuated to grid 66 KV, Gujarat
- Design Consultancy for 5 MW, 33 and 66 KVA evacuation SPV power plant – 2 projects, Gujarat.
- Rooftop 100–200 KW Power plant, Grid tied, 3 phase – 9 nos.
- 10 MW, 66KV – under execution
- Solar power plant for telecom towers between 2 KW to 10 KW.
- Village electrification – 5 KW to 40 KW

Team – Major Clients direct and indirect for solar and other areas

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|-----------------------------------|--|
| ‣ MEMC / SunEdison, USA | ‣ Kumar Builders |
| ‣ BHEL | ‣ Brigade Group / builders |
| ‣ AES, USA | ‣ Asian Paints |
| ‣ Enfinity, Belgium | ‣ ICICI Manipal |
| ‣ Birla Power | ‣ Gujarat Power corporation |
| ‣ TATA Motors | ‣ Karnataka Power corporation |
| ‣ Vodafone | |
| ‣ Euro Ceramics | |
| ‣ Kirloskar | |
| ‣ Fuji Films | |
| ‣ Symbiosis Group of Institutions | Etc. Etc. amongst several other distinguished clientele for PV and Thermal systems |
| ‣ Sinhagad Group of Institutions | |
| ‣ Tolani Maritime Institute | |

Solar PV – Grid system, Gujarat / Karnataka



PV array, Control Room, Inverter Room



Site construction sequence, – clearing, levelling,



Inverter room façade, column, base plate, frame mounting



Module mounting, clamp fixing, cabling interconnection, trays, night work, module cleaning

Solar Power Plant, Crystalline Array , Thin Film Array in Gujarat



View of 5 MW power plant with crystalline panels, 66 KV grid supply